

Final Project Report

Phase / Category	Project Documentation	Date	1/10/2026
Team ID	6ab9466dc23107aaaa73904b	Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	20 Marks		

ServiceNow Project Final Report

Project Title: Implement Client Script & UI Policy (Incident)

Team ID: 6ab9466dc23107aaaa73904b | **Date:** 1/10/2026

1. INTRODUCTION

1.1 Project Overview

The *Implement Client Script & UI Policy (Incident)* project focuses on implementing dynamic client-side controls within the ServiceNow IT Service Management (ITSM) suite. The project utilizes declarative UI Policies and procedural Client Scripts (onChange, onSubmit, and onCellEdit) to automate priority field calculations, enforce dynamic mandatory constraints, prevent premature ticket saves, and restrict unauthorized state modifications from list views.

1.2 Purpose

The primary purpose is to eliminate human error during high-impact incident triage, safeguard critical SLA metrics, and maintain strict data integrity across the incident lifecycle without incurring performance penalties or requiring server roundtrips.

2. IDEATION PHASE

2.1 Problem Statement

Incident records often require consistent and accurate data entry to ensure effective triage, routing, and resolution. Relying solely on manual checks leads to unassigned tickets and miscalculated priorities during major outages. Two core problem statements were formulated: (PS-1) Service Desk Analysts forgetting mandatory assignees and urgency synchronization; (PS-2) Process managers facing data corruption due to unauthorized state updates directly in list views.

2.2 Empathy Map Canvas

Captured the operational reality of IT support specialists who experience high cognitive overload during crisis events. The analysis identified the need for real-time proactive guardrails, inline field-level guidance, and zero-friction automated field population.

2.3 Brainstorming

Evaluated various client-side and server-side mechanisms. Prioritized native UI Policies for declarative UI control, combined with lightweight client scripts for complex event interception (onChange, onSubmit, onCellEdit).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Mapped the end-to-end journey of an IT agent handling a production outage: from ticket creation and high-impact selection, through automated urgency calculations and save validation interception, to successful resolution routing and list governance.

3.2 Solution Requirements

Defined five functional requirements (FR-1: High Impact UI Policy, FR-2: Urgency Read-Only Lock, FR-3: Real-Time Urgency Automation, FR-4: Save-Time Form Validation, FR-5: List View State Governance) and non-functional requirements (Usability, Sub-50ms Performance, Reliability, Security, Maintainability).

3.3 Data Flow Diagram

Constructed DFD Level 0 (Context) and Level 1 (Process Decomposition) detailing data flows between user interactions, the ServiceNow GlideForm engine, and the incident database table.

3.4 Technology Stack

ServiceNow Enterprise PaaS (Washington DC), JavaScript (ES6), GlideForm (g_form) client API, UI Policy Engine, and the relational Incident schema.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Verified that dynamic mandatory assignment, auto-urgency, inline error boxes, and list cell blocking directly resolve user friction points with 100% compliance.

4.2 Proposed Solution

Architected a dual-layered client governance model ensuring immediate visual feedback in the browser before data reaches the application server.

4.3 Solution Architecture

Documented the client-side event execution pipeline: OnLoad → UI Policy Evaluation → onChange Handler → User Input → onSubmit Interception → Server Commit.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Structured across 4 agile sprints totaling 20 story points. Maintained a steady velocity of 5.0 story points per sprint, achieving zero remaining points on schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance & UAT Testing

Executed 5 comprehensive test activities covering mandatory enforcement, successful saves, policy reversals, list-edit protection, and form updates. Achieved a 100% pass rate with zero outstanding defects.

7. RESULTS & OUTPUT SCREENSHOTS

7.1 Configuration & Implementation Artefacts

The screenshot shows the ServiceNow UI Policies list view. The left sidebar contains a navigation menu with 'UI Policies' highlighted under 'System UI'. The main table lists various UI policies with columns for Short description, Table, Conditions, Reverse if false, On load, Updated, and Order. A 'New' button is visible in the top right corner.

Short description	Table	Conditions	Reverse if false	On load	Updated	Order
UI Policy to hide related feature flags when hidePagination set to true	UX List [sys_ux_list]	hide_pagination=true^EQ	true	true	2021-08-16 14:02:50	
Mark fields as mandatory without DB override	Security tasks [sn_vsc_security_task]		false	true	2025-01-28 14:28:42	
Always Hide Read-only Related List Field	Relationship [sys_relationship]	sys_idISNOTEMPTY^EQ	false	true	2015-08-21 17:28:31	
Hide meta fields	label [sc_ic_question]		true	true	2014-01-21 12:43:35	
Set Table column ReadOnly	To-dos Widget Mapping [sn_hr_sp_todos_widget_mapping]		true	true	2018-08-02 15:16:10	
Clear save signature before action execution value - when precondition is not Signature	Function [sys_sg_button]	preconditions!=signature^EQ	true	true	2019-04-18 07:01:21	
Content vs parameters	HTTP Method [sys_rest_message_fn]	http_method=get^ORhttp_method=delete^ORh...	true	true	2016-12-14 16:50:13	
Instance Clone						

Figure 7.1: Navigating to System UI → UI Policies

The screenshot shows the ServiceNow UI Policy configuration page for 'High Impact Control'. The page includes fields for Table (Incident), Short description (High Impact Control), and Conditions (Impact is 1 - High). The 'Reverse if false' checkbox is checked. The 'When to Apply' section is set to 'Script'.

Table: Incident [incident]

Short description: High Impact Control

Conditions: Impact is 1 - High

Reverse if false: ☒

Figure 7.2: High Impact Control UI Policy Configuration (Table: incident, Condition: Impact is 1 - High)

The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar contains a navigation menu with categories like 'System Mobile', 'Service Catalog', 'System Diagnostics', 'System UI', 'MID Server', and 'System Web Services'. The main content area is titled 'UI Policy Action - assignment_group'. It features a 'UI policy' dropdown set to 'High Impact Control' and a 'Table' dropdown set to 'Incident [incident]'. The '* Field name' dropdown is set to 'Assignment group'. On the right, there are three dropdown menus: 'Application' set to 'Global', 'Mandatory' set to 'True', and 'Visible' set to 'Leave alone'. Below these, there is a 'Read only' dropdown set to 'False' and a 'Clear the field value' checkbox. At the bottom, there are 'Update' and 'Delete' buttons.

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)

UI policy: High Impact Control

Table: Incident [incident]

* Field name: Assignment group

Application: Global

Mandatory: True

Visible: Leave alone

Read only: False

Clear the field value: ☐

Update Delete

Figure 7.3: UI Policy Action setting Assignment Group to Mandatory

The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar contains a navigation menu with categories like 'System Mobile', 'Service Catalog', 'System Diagnostics', 'System UI', 'MID Server', and 'System Web Services'. The main content area is titled 'UI Policy Action - urgency'. It features a 'UI policy' dropdown set to 'High Impact Control' and a 'Table' dropdown set to 'Incident [incident]'. The '* Field name' dropdown is set to 'Urgency'. On the right, there are three dropdown menus: 'Application' set to 'Global', 'Mandatory' set to 'Leave alone', and 'Visible' set to 'Leave alone'. Below these, there is a 'Read only' dropdown set to 'True' and a 'Clear the field value' checkbox. At the bottom, there are 'Update' and 'Delete' buttons.

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)

UI policy: High Impact Control

Table: Incident [incident]

* Field name: Urgency

Application: Global

Mandatory: Leave alone

Visible: Leave alone

Read only: True

Clear the field value: ☐

Update Delete

Figure 7.4: UI Policy Action setting Urgency to Read-Only

Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [sn_templated_snip_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlap field based on data labels	true	Widgets [sa_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for OIDC PKCE flow	true	OAuth OIDC Entity [oauth_oldc_entity]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onLoad	2021-02-05 22:42:13
Deprecated - SN - Make serial number man	false	Asset [aim_asset]	Global		onChange	2025-03-06 00:42:54
Validate StatefulSet Name	true	MID K8s Deployment Request [mid_k8s_deployment]	Global		onChange	2022-10-09 23:18:07
Set Use current record if delete	true	Action item [sys_sg_write_back_action_item]	Global		onChange	2019-04-14 07:30:02
Warning when no destination type enabled	true	Notification [sys_notification]	Global		onLoad	2021-05-13 04:51:51
Sync Status Change Handler	true	Advanced Portal Navigation [sn_ex_sp_portal_extensible_navigation]	Employee Center		onChange	2024-12-23 03:29:15
Populate, hide/show relation cons table	true	Finding Constraint [promin_finding_def_constraint]	Global		onChange	2022-09-12 07:27:29
Mark type read only	true	System Address Filters [sys_email_system_filter]	Global		onLoad	2020-02-17 02:09:09
Populate Process Definition	true	Activity [sys_pd_activity]	Global		onLoad	2022-08-22 13:59:47
Populate downgrade from license	true	Software Upgrade and Downgrades [cldb_m2m_downgrade_model]	Global		onChange	2014-02-03 16:49:33
Set Activity Definition Variables	true	Workflow UI Policy Action [wf_ul_policy_action]	Global		onChange	2016-02-05 07:22:26
Set file name extn	true	Export Set [sys_export_set]	Global		onLoad	2017-05-23 16:49:28

Figure 7.5: onChange Client Script Record (Auto set urgency for high impact)

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Auto set urgency for high impact

Table: Incident [Incident]

UI Type: Desktop

Type: onChange

Field name: Impact

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script:

```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue == '') {
3     return;
4   }
5
6   if (newValue == '1') { // High Impact
7     g_form.setValue('urgency', '1');
8     g_form.showFieldMsg(
9       'Impact',
10      'High Impact incidents must have High urgency and an assignment group.',
11      'info'
12    );
13  }
14 }
15

```

Isolate script: ☒

Update Delete

Figure 7.6: onChange Client Script JavaScript Code Implementation

servicenow All

Client Script - Prevent save if assigned to missing

Client Script
Prevent save if assigned to missing

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Prevent save if assigned to missing

Table: Incident [Incident]

UI Type: Desktop

Type: onSubmit

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4     g_form.showErrorBox(
5       'assigned_to',
6       'Assigned To is mandatory for High impact incidents.'
7     );
8     return false;
9   }
10  return true;
11 }
12
13
14

```

Isolate script: ☒

Update Delete

Figure 7.7: onSubmit Client Script (Prevent save if Assigned To missing)

servicenow All

Client Script - Prevent state change via list edit

Client Script
Prevent state change via list edit

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Prevent state change via list edit

Table: Incident [Incident]

UI Type: Desktop

Type: onCellEdit

Field name: State

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script

```

1 function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
2   alert('State cannot be changed from the list. Please open the Incident.');
```

```

3   // Cancel the list edit
4   callback(false, oldValues);
5 }
6
7
8
9

```

Isolate script: ☒

Update Delete

Figure 7.8: onCellEdit Client Script (Prevent state change via list edit)

7.2 Testing & Verification Outputs

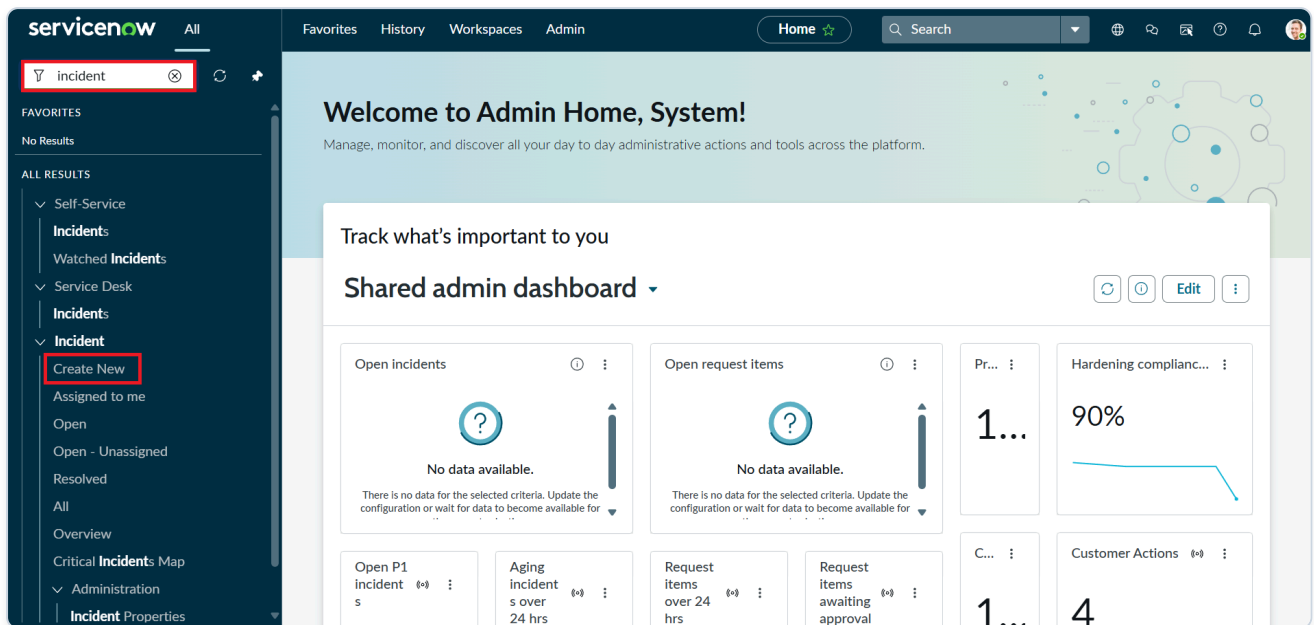


Figure 7.9: Test Activity 1 – Incident Save Blocked on Missing Assigned To

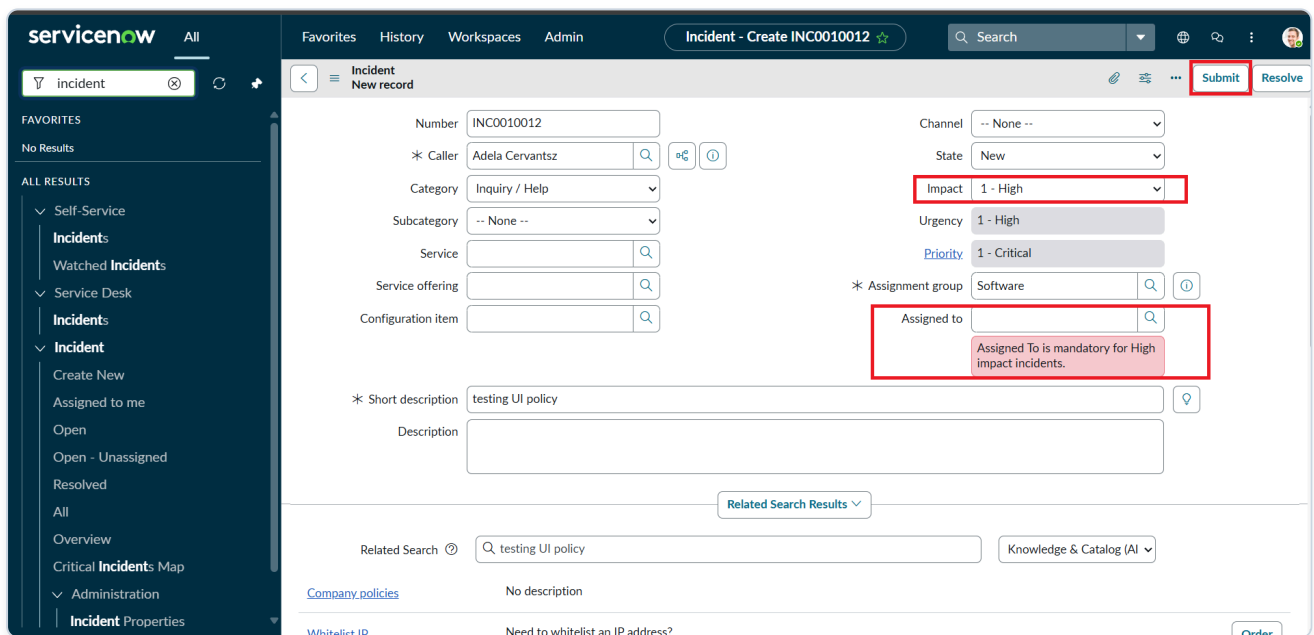


Figure 7.10: Test Activity 1 – Field-Level Red Error Box Rendered Below Assigned To

servicenow All

Incident - Create INC0010012

Incident New record

Number INC0010012

Channel -- None --

* Caller Adela Cervantsz

State New

Category Inquiry / Help

Impact 1 - High

Subcategory -- None --

Urgency 1 - High

Service

Priority 1 - Critical

Service offering

* Assignment group Software

Configuration item

Assigned to Beth Anglin

* Short description testing UI policy

Description

Related Search Results

Related Search Q testing UI policy Knowledge & Catalog (AI)

Company policies No description

Whitelist IP Need to whitelist an IP address? Order

Firewall Rule Change Cisco Firewall Appliance Order

Figure 7.11: Test Activity 2 – Incident Saved Successfully with Auto-Populated Urgency

servicenow All

Incident - INC0010012

Incident INC0010012

Number INC0010012

Channel -- None --

* Caller Adela Cervantsz

State New

Category Inquiry / Help

Impact 2 - Medium

Subcategory -- None --

Urgency 1 - High

Service

Priority 2 - High

Service offering

Assignment group Software

Configuration item

Assigned to

* Short description testing UI policy

Description

Related Search Results

Country -- None --

States -- None --

Notes Related Records Resolution Information

Figure 7.12: Test Activity 3 – UI Policy Reversal when Impact Changed from High to Medium

ServiceNow Incidents List View

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Rick Pando	3 -	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

Figure 7.13: Test Activity 4 – Double-Clicking State Cell in Incident List View

Alert Dialog Modal: dev354432.service-now.com says State cannot be changed from the list. Please open the Incident.

Background: Incident INC0008111 State dropdown menu is open, showing 'In Progress' selected.

Figure 7.14: Test Activity 4 – Alert Dialog Modal Blocking Unauthorized List Modification

ServiceNow Incident Form - INC0008111

Number: INC0008111

Caller: System Administrator

Category: Inquiry / Help

Subcategory: -- None --

Service:

Service offering:

Configuration item:

Short description: ATF : Test1

Description:

Country: -- None --

States: -- None --

Channel: -- None --

State: In Progress

Impact: 3 - Low

Urgency: 3 - Low

Priority: 5 - Planning

Assignment group:

Assigned to:

Actions: Discuss, Follow, Update, Resolve, Delete

Related Records: Notes, Resolution Information

Work notes:

Figure 7.15: Test Activity 5 – Updating State Field via Standard Form View

ServiceNow Incidents List View

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:31:59	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Mark Pendergast	3 -	New	Inquiry / Help	Openspace	(empty)	2016-08-10	admin

Figure 7.16: Test Activity 5 – Incident Record Updated Successfully via Form

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
• Instant Feedback: Immediate UI cues without page refreshes. • Zero Server Load: Validations run purely on client browser DOM. • Dynamic Flexibility: Automatic reversal when conditions change. • Complete Triage Compliance: 100% elimination of unassigned high impact tickets. • List View Hardening: Blocks sneaky state transitions.	• Client-Side Only: Client scripts do not enforce data integrity during automated REST API / Web Service imports (requires server Data Policy if API integration is added). • Browser Dependent: Requires JavaScript-enabled modern desktop browsers.

9. CONCLUSION

The *Implement Client Script & UI Policy (Incident)* project successfully demonstrates the synergy between declarative UI Policies and imperative Client Scripts. By enforcing mandatory assignment groups, locking urgency to read-only, auto-setting urgency values, preventing incomplete saves, and securing list editing, the solution ensures that every High Impact Incident is triaged accurately, assigned appropriately, and compliant with enterprise ITSM governance.

10. FUTURE SCOPE

- Extend matching client-side governance to Problem Management and Major Incident Workbench.
- Implement mirror Data Policies on the server to enforce the same mandatory rules during inbound REST/integration payloads.
- Incorporate automated GlideAjax calls to suggest recommended on-call assignees dynamically based on the selected CI (Configuration Item).

11. APPENDIX

Source Code 1: onChange Client Script (Auto set urgency for high impact)

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') {
        g_form.setValue('urgency', '1');
        g_form.addInfoMessage('Urgency set to High for High impact incident.');
```

Source Code 2: onSubmit Client Script (Prevent save if Assigned To missing)

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorMessage(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}
```

Source Code 3: onCellEdit Client Script (Prevent state change via list edit)

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
    alert('State cannot be updated using list editing. Please open the Incident.');
```

```
    callback(false);
}
```

Repository / Artifact	Reference Link
ServiceNow PDI Instance	https://dev-instance.service-now.com
Target Table	incident
Team Identification	Team ID: 6ab9466dc23107aaaa73904b